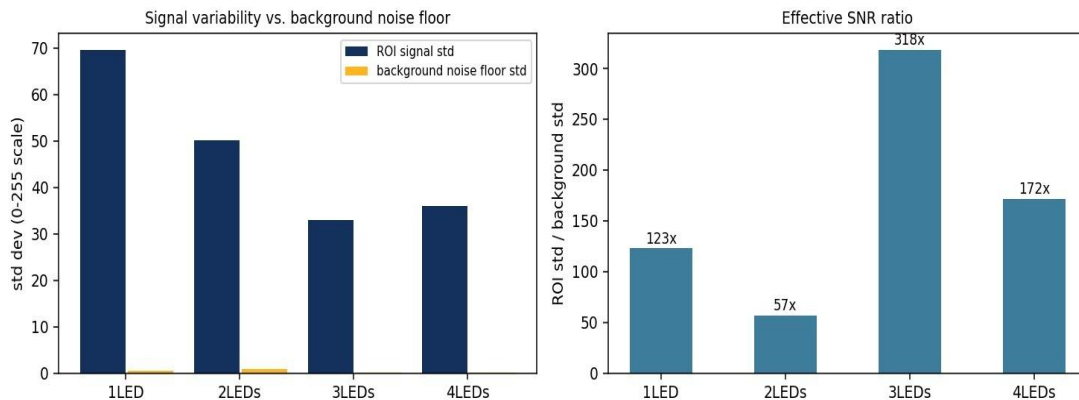


Frame & Intensity EDA — Signal Plots

1. SETUP

Using the Day 9 frozen ROI boxes (fixed per video, justified by Day 8's sub-5px motion measurement) and the Day 7 active-window timestamps. Expected symbol timing: 92 bps $\rightarrow$ 10.87 ms bit period; 260 fps camera $\rightarrow$ 3.846 ms frame period $\rightarrow$ 2.826 frames per bit.

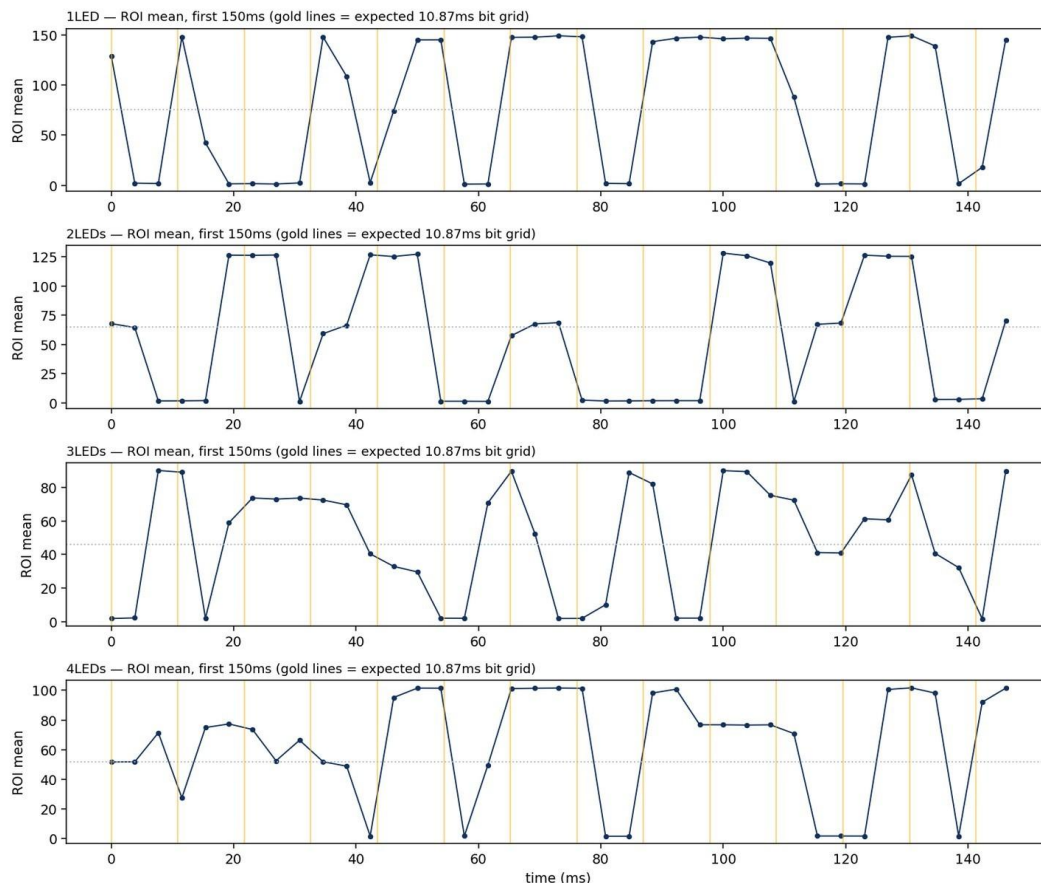
2. SIGNAL QUALITY: ROI VS. BACKGROUND NOISE



All four ROIs show strong signal-to-noise — 57x to 318x the background noise floor. Signal quality is not a limiting factor for any of the four videos; decoding difficulty, if any, will come from timing, not from weak or noisy signal.

3. ZOOMED SIGNAL VS. EXPECTED BIT-PERIOD GRID

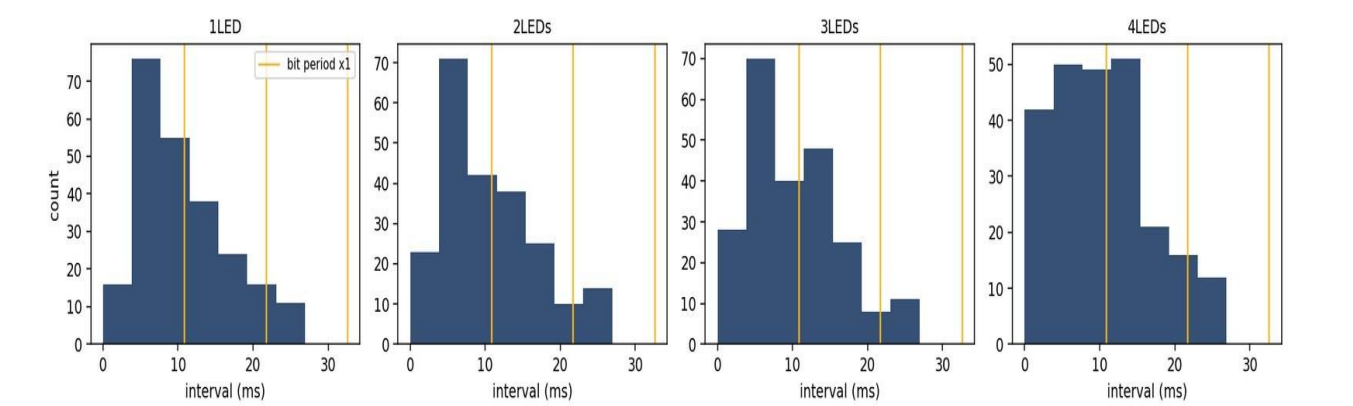
First 150ms of each ROI intensity trace. Gold lines mark where bit boundaries would fall under a naive assumption of a clean 10.87ms clock starting at frame 0.



Plateau widths vary considerably — consistent with runs of repeated bits under NRZ-OOK encoding — but do not line up cleanly with the naive fixed grid, which assumed a known start phase and perfectly stable clock. Expected, not a data quality issue.

4. TRANSITION TIMING VS. THEORETICAL SYMBOL PERIOD

Threshold-crossing transitions were detected on the raw per-frame ROI mean (global midpoint threshold) and the time between consecutive transitions measured.



File	Transition Interval Distribution (percentiles, in units of frame-period)
1LED_92bps.mp4	10th–90th pct: 2x, 2x, 2x, 4x, 6x frame periods
2LEDs_92bps.mp4	10th–90th pct: 1x, 2x, 2x, 4x, 6x frame periods
3LEDs_92bps.mp4	10th–90th pct: 1x, 2x, 2x, 4x, 5.8x frame periods
4LEDs_92bps.mp4	10th–90th pct: 1x, 2x, 3x, 4x, 6x frame periods

Key finding: intervals cluster at integer multiples of the 3.846ms camera frame period, not at multiples of the true 10.87ms bit period. A naive global-threshold crossing on the per-frame mean is dominated by frame-grid quantization, not the real symbol clock — confirming the sync risk predicted from the non-integer frames-per bit ratio on Day 9. Reliable decoding will need sub-frame/rolling-shutter extraction or explicit clock recovery, not just a smarter threshold.

5. AMBIENT LIGHT / DRIFT — SHORT TIMESCALE

Day 8 checked ambient drift across each video's full duration. This checks the ROI's own 'off-state' baseline within the 3-second window relevant to decoding.

File	Off-State Baseline	Std Dev	Linear Drift
1LED_92bps.mp4	2.56	3.15	+0.022 /s
2LEDs_92bps.mp4	2.85	2.62	+0.439 /s
3LEDs_92bps.mp4	2.52	1.88	+0.018 /s
4LEDs_92bps.mp4	2.20	2.41	−0.264 /s

Negligible drift in all four videos at this timescale too — consistent with the Day 8 full-duration finding. Ambient light is not a concern for decoding.

6. SUMMARY

Question (Day 10 task)	Finding
Frame/intensity plots	Extracted for all 4 videos; strong, visually distinct on/off structure
ROI statistics & noise	SNR 57x–318x across all 4 — excellent signal quality, not a limiting factor
Transitions & symbol period	Threshold-crossing intervals are frame-quantized, not bit-period-aligned — confirms Day 9's sync risk is real
Ambient light / drift	Negligible at both full-video (Day 8) and 3s-window (Day 10) timescales

7. NEXT STEP

Day 11 (Signal Preprocessing) implements the ROI extraction and smoothing pipeline properly and begins sub-frame timing recovery to address the finding in Section 4, this is now the primary technical challenge, not a data-quality gap.